

ALIMA BAZENOVA

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ML engineer focused on **computer vision** for scientific and medical imaging

EDUCATION

Eötvös Loránd University, Budapest, Hungary

September 2024 – June 2026

MSc in Environmental Sciences

CGPA: 4.9/5.0

Thesis: Automated Pollen Detection using CNN and Transformer Detectors, and Younger Dryas Climate Reconstruction

Nazarbayev University, Astana, Kazakhstan

August 2020 – May 2024

BSc in Biological Sciences

CGPA: 3.6/4.0

RESEARCH EXPERIENCE

Research Assistant – AI for Clinical CT Imaging

August 2026 – Present

School of Medicine, Nazarbayev University (PI: Dr. Srinivasa Bolla)

- Developed a deep-learning pipeline for opportunistic osteoporosis and sarcopenia screening from routine lumbar-spine CT, linking L3 psoas muscle biomarkers to DXA bone mineral density in 280 patients.
- Automated vertebra and psoas segmentation on a multi-vendor cohort (5 manufacturers): baseline mean Dice 0.89 vs expert-corrected masks; the validated annotation set is used to fine-tune the model and train CNN classifiers for osteoporosis prediction.

Research Assistant – Automated Microscopy & Image Analysis

September 2025 – July 2026

Palaeoecology Research Group, Eötvös Loránd University (PI: Dr. Enikő Magyari)

- Engineered an automated microscope-scanning and live-detection workflow replacing manual slide counting (75% faster); relative abundances matched an expert palynologist on 3 fully independent slides (Pearson $r = 0.86$, Spearman $\rho = 0.80$).
- Built and released a 22-class detection dataset (3,001 images, 8,460 boxes) and benchmarked 7 CNN/transformer detectors (YOLOv8-v12, YOLO26, RT-DETR; 21 runs) under a leakage-free depth-based split: best YOLOv12l reached 0.496 pollen mAP@0.5, while a random split inflated it to 0.749 (+51%).
- Trained a ResNet50 classifier for subfossil chironomid identification (evaluated in ParticleTrieur) on 27,774 specimen images from 57 lakes in 5 countries.

INDUSTRY EXPERIENCE

Junior ML Engineer – Video Behaviour Recognition

May 2025 – August 2025

Green Eye Ltd

- Built a 6-class behaviour dataset (~20K labelled frames/clips); benchmarked 10+ CV approaches and pretrained-CNN transfer-learning baselines (15+ configurations) by precision, recall, macro F1 and confusion matrices.

PROJECTS

Safe Camera – Multi-Stream Video Behaviour Recognition

2026

- Fine-tuned Kinetics-400 pretrained R3D-18 on RWF-2000 (74.5% acc., 80.5% recall, 75.9% F1); K-of-N temporal smoothing with a calibrated threshold cut false detections 4× at fixed recall (44.2 → 10.7 per camera-hour, 2.06 h of footage).
- Processed 8 synchronised streams at 25 FPS with 0.2% frame loss and 176 ms p95 latency (bounded drop-oldest queues, auto-reconnect, stall watchdog).
- Cut inference cost 3.8× (658 → 175 ms/window), accuracy-neutral by McNemar test; ONNX Runtime and int8 quantisation reported as negative results.

Health Passport AI – Big Berlin Hackathon

April 2026

- Gemini-API pipeline that parses PDF/DOCX/image medical records, translates them and flags abnormal lab values.

SKILLS

Computational	Deep learning for image/video (PyTorch): object detection, CT segmentation, pose estimation, action recognition; dataset construction & annotation QC; real-time multi-stream video
Quantitative	Statistical testing (ANOVA, non-parametric), inter-observer agreement, leakage-free validation; precision/recall/F1/mAP/Dice
Programming	Python (NumPy, pandas, OpenCV, SimpleITK, nibabel), Git, FastAPI

PUBLICATIONS

Khamkhash, L., **Bazanova, A.**, Rustembekkyzy, K., Balarabe, B. Y. Y., & Atabaev, T. S. (2025). Microwave-assisted synthesis of Gr-ZnO nanorods for solar photocatalytic removal of methylene blue. *Materials Advances*, 6, 9495–9504. doi:10.1039/D5MA00612K